

FACULTY OF SCIENCE
COMPUTER & INFORMATION SCIENCE DEPARTMENT
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

BSC Computer	Computer Science	Mathematics	Common Courses	Total
Four-year Programme	43	5	4	52
Three-year Programme	34	3	3	40

Course structure

THREE YEAR PROGRAMME

YEAR 1

Semester I

GS 100	Christian Ethics	2
MAT 101	Differential Calculus	3
CMT 109	Database systems	3
CMT 203	Introduction to Systems administration	3
CMT 205	Computer Architecture	3
CMT 208	Introduction to Artificial Intelligence	3
CMT 212	Cloud Computing Essentials	3
CMT 210	Object Oriented Programming I	5

Semester II

MAT 200	Integral Calculus	3
MAT 262	Probability & Statistics	3
PHY 205	Digital Computer Electronics	3
CMT 206	Operating Systems	3
CMT 209	Web Development	5
CMT 211	Object Oriented Programming II	5

YEAR 2, Semester I

GS 103	Introduction to the Bible	2
GS 104	Environmental Studies	2
CMT 300	Data Communications	3
CMT 302	Advanced Database Systems	3
CMT 304	Data Structures & Algorithms	5
CMT 308	Distributed Systems	3
CMT 310	Object Oriented System Analysis and Design	3
CMT 314	Mobile Application Development	5

YEAR 2, semester II

Core Courses

CMT 301	Research Methodology	3
CMT 307	.NET Programming	5
CMT 317	Information Systems Security	3
CMT 311	Fundamentals of Software Engineering	3
CMT 313	Computer graphics	3
CMT 316	Human Computer Interface	3
CMT 320	Industrial Attachment	N/A

YEAR 3, Semester I

Core Courses

CMT 400	Research Project	3
CMT 409	Introduction to Data Science	3
CMT 408	Software Project Management	3

Select **THREE** courses from **ONE** of the areas of specialization.

(a) Data Science

CMT 426	Programming Using Scala	3
CMT 428	Hadoop Platform and Application Framework	3
CMT 432	Introduction to Apache Spark	3
CMT 434	Python for Data Science	3

(b) Distributed Systems

CMT 412	Distributed Operating Systems	3
CMT 447	Web Application Security	3
CMT 448	Distributed Systems Programming	3

(c) IT security

CMT 436	Security and Privacy for Big Data /Cloud Security	3
CMT 438	Ethical Hacking	3
CMT 447	Web Application Security	3

d)Artificial Intelligence

CMT 432	Introduction to Apache Spark	3
CMT 434	Python for Data Science	3
CMT 440	Advanced Artificial Intelligence	3
CMT 443	Data Mining	3

YEAR 2, SEMESTER II

Core Courses

CMT 400	Research Project	3
CMT 403	Information Systems Management	3
CMT 407	Social Informatics	3

Electives Courses

Select THREE courses from ONE of the areas of specialization.

(a) Data Science

CMT 437	Big Data Analysis	3
CMT 444	Distributed Machine Learning	3
CMT 445	Business intelligence	3
CMT 448	Data analytics Essentials	3

(b) Distributed Systems

CMT 410	Network Programming	3
CMT 413	Network Design and Administration	3
CMT 417	Advanced Web Development	3
CMT 446	Information Systems Audit	3

(c) IT security

CMT 446	Information System Audit	3
CMT 449	Principles of Cryptography	3
CMT 451	Mobile Security	3

(d) Artificial Intelligence

CMT 444	Distributed Machine Learning	3
CMT 441	Logic Programming	3
CMT 442	Neural Networks	3
CMT 445	Business Intelligence	3